Parametric LCR circuit with C(t)

Does a pumped capacitor draw energy "from the time domain", or from the pump? — register entry H16

kronspice simulation report · 2026-11-07

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Summary

F. D. Amador and H. Eckardt ("Simulation of a Parametric Resonance Circuit", A.I.A.S./UPITEC, 30 March 2012) drive a series R, L, C(t) loop and report a coefficient of performance (COP) of about 200 against the source's input, read as energy "from the time domain" (spin connection resonance). This example reproduces every case in the paper with kronspice's time-varying capacitor, and puts on the ledger the one agent the paper's energy sum leaves out: the pump that varies C. The paper's reading is hypothesis H16; the standard prediction is parametric excitation (Mandelstam & Papalexi 1932; Kouřil & Vrba §10.2).

- **Every case is reproduced**, and in every one the source plus the pump pays for the heat in R and the stored energy; the books close to 0.001–0.4 % where resolved.
- **Tuned pumping at $2\omega_0$ grows without limit** (87.3 A at 30 ms, σ = 245 1/s, as Kouřil & Vrba predict), paid by the pump: 256 J in against −0.085 J from the source.
- **Detuned to 0.97 f_0 at φ = −90°**, the case the paper highlights: bounded 16.1 A and 2.43 kW; the source gives 6.75 J, the pump 1139 J, R takes 1143 J.
- **The paper's COP divides heat in R by the source energy alone:** 169 for that case. Divided by source plus pump it is 0.998 in the bounded cases and below 1 everywhere.
- **Pumping at the drive frequency** (§3.2) grows only for a_0 of about 0.7 or more, as the second Mathieu tongue requires.
- **Bench test:** measure the power that drives C(t) — the motor shaft or the varactor bias supply.

What is being tested

Hypothesis H16 (Amador & Eckardt 2012): a parametric LCR circuit with C(t) draws energy "from the time domain", with a COP of about 200 by source input. **Standard prediction beside it:** a capacitor whose value is varied is driven by an outside agent (the motor that moves the plates, or a varactor's bias supply), and that agent does work on the circuit, $P_{pump} = -\dot{s}\cdot\tfrac{1}{2}\cdot C\cdot v^2$. Parametric excitation (Mandelstam and Papalexi 1932, the paper's ref. [13]; Kouřil & Vrba §10.2.2) says the circuit self-excites when the elastance modulation depth m exceeds a threshold, and that all its energy comes from the source and the pump.

The circuit

Paper Fig. 2, eqs. 19–27: a sine source $U = U_0\cdot\sin(\omega t)$, inductance L, resistance R and a capacitance $C_{eff}(t) = C\cdot(1 + a_0\cdot\sin(\omega_1\cdot t + \varphi))$, with $q = C_{eff}\cdot u_C$ (eq. 25). kronspice's VaryingCapacitor, $q = s(t)\cdot C\cdot v$, is eq. (25) exactly. Every case is checked as source + pump = heat in R + change of stored energy.

parameter (paper eqs. 28–36)	value
U_0	5 V
C, L, R	230 nF, 20 mH, 20 Ω
resonance f_0	2346.6 Hz
quality factor Q	14.74
damping α = R/2L	500 1/s
pump	$\omega_1 = 2\omega_0$, $a_0 = 0.2$, φ = 0

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Standard theory beside the paper

- $a_0 = 0.2$ gives an elastance depth m = 0.2020 against the tuned self-excitation threshold $m_c = 2/Q = 0.1356$ (K&V 10.101), so pumping at $2\omega_0$ self-excites.
- $C(1 + a \sin)$ raises the mean elastance by $1/\sqrt{(1 - a^2)}$, so the circuit's own frequency is 1.0103 f_0.
- Pumped at $2 \times 0.97\ f_0$ the detuning is 4.2 % and the threshold rises to 0.2144 > m: below self-excitation, the circuit is a parametric amplifier of the drive (K&V 10.103).

Results

"Books" is the closure of source + pump against heat in R + stored energy.

paper case	kronspice	books
Fig. 4, constant C at f_0	peak 0.25 A = U_0/R; source 0.0175 J → R 0.0169 J + stored 0.0006 J	0.001 %
§3.1 Fig. 5, $C(1 + 0.2 \sin 2\omega_0 t)$ at f_0, 30 ms	87.3 A and growing without limit, σ = 245 1/s as K&V predict; source −0.085 J, pump 256 J → R 165 J + stored 91 J	0.07 %
§3.1 Figs. 6–7, drive and pump at 0.97 f_0, φ = 0	bounded 7.34 A, 504 W; pump 241.5 J, source −3.8 J	0.04 %
same, φ = −90°	bounded 16.1 A, 2.43 kW; source 6.75 J, pump 1139 J → R 1143 J	0.04 %
same, φ = +90°	bounded 5.72 A, 307 W; pump 138 J	0.04 %
§3.2 Fig. 8, $C(1 + 0.2 \sin^7 \omega_1 t)$, ω_1 = drive = 1.07 ω_0	0.123 A, no growth	0.4 %
§3.2 Fig. 8, $a_0 = 0.8$	207 A; pump 607 J → R 262 J + stored 345 J	0.3 %
§3.2 Fig. 9, rectangular C 0.8C ↔ 1.2C at 0.95 f_0, drive off after 2 periods	decays, 1.1 mA	0.4 %
§3.2 Fig. 9, rectangular C 0.2C ↔ 1.8C	7.6 kA from a 2-period drive, or 443 A from a 1 mA start with no drive; pump 2360 J → R 614 J + stored 1854 J	few % (integration error: 0.97 % → 0.32 % from 2000 to 8000 steps per period)

The COP, with and without the pump

- The paper's COP divides the heat in R by the source energy alone: 0.964 for Fig. 4, 169 for the φ = −90° case the paper highlights, and negative where the source absorbs.
- Divided by source plus pump it is 0.998 in the bounded cases and below 1 everywhere.
- In every case the heat in R and the stored energy are paid by the source plus the pump that varies C. The pump's share is what the paper's "energy from spacetime" counts.

Ledger and validation

- The pump's power $P_{pump} = -\dot{s} \cdot \tfrac{1}{2} \cdot C \cdot v^2$ is on the ledger in every run; closure 0.001 % to 0.4 %, and a few % only for the 0.2C ↔ 1.8C rectangular case, where it falls with the step as integration error does.
- Tests (main_test.go): classical amplitude (TestClassical), tuned growth paid by the pump (TestTunedGrowth), detuned bounded case (TestDetunedBounded), second tongue $a_0 = 0.2$ against 0.8 (TestSecondTongue).
- Export sheet, Fig. 4 constant-C case at 2346.61 Hz: 0.25 A through R (5 V across), MNA and Kron's torn solve agree to 1.3×10^{-14} of the largest node voltage, and the power taken by the R, L, C branches is the same torn and untorn (Roth 1959 §5). The pumped-C circuit is nonlinear, so its sheet gives MNA only.

Limits of the model

- The capacitor variation s(t) is imposed; the pump is ideal, and only its power on the circuit is booked, not the losses of a real motor or bias supply.
- Lumped series R, L, C; no losses beyond R.

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Status

H16: untested (reproduced; the COP omits the pump). All of the paper's cases are reproduced: $U_0/R = 0.25$ A classical case; unbounded growth at $2\omega_0$ pumping ($\sigma = 245$ 1/s); bounded ≈ 16 A, 2.4 kW at 0.97 f_0, $\varphi = -90°$ (source-only ratio 169); pumping at ω_0 grows only for a_0 of about 0.7 or more. In every case the source plus the pump pays for R and the stored energy. The bench test is to measure the power that drives C(t), at the motor shaft or the varactor bias supply.

Reproducing the results

```
go run ./examples/parametric_lcr
go test ./examples/parametric_lcr
KRONSPICE_EXPORT=1 go test ./examples/parametric_lcr -run TestExport   # export/ sheets
```

Related: the same parametric pumping in Kron's machine frame, examples/kron_machine and examples/parametric_pump.

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